

Chunfan Yao

UNDERGRADUATE RESEARCHER · AI & ALGORITHMS

Berkeley, CA (Exchange) / Hong Kong

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“Building benchmarks and agents for hard combinatorial reasoning.”

Summary

Rising junior in Artificial Intelligence at HKUST, currently on exchange at UC Berkeley (Fall 2026). Co-first author on a paper in preparation for *Physical Review X Quantum* that constructs the first unit-weight gadgets for triangular Rydberg lattices, verified by 5.6 billion exhaustive lattice-animal enumerations. As a key contributor to an LLM bug-finding benchmark ([problem-reductions-benchmark](#)), designed the evaluation framework that measures how efficiently language models locate logical errors in combinatorial reduction code. Seeking research opportunities at the intersection of LLM reasoning, agentic systems, and combinatorial problem-solving.

Research Experience

Quantum Computing & Graph Optimization Lab — Prof. Jinguo Liu, HKUST

Hong Kong

UNDERGRADUATE RESEARCH ASSISTANT (Co-First Author)

2025 – Present

- Co-authoring a paper (in preparation for *PRX Quantum*) proving the existence of the first unit-weight CROSS and CROSS+EDGE gadgets for triangular Rydberg lattices, enabling universal MIS encoding under a single uniform detuning — closing an open problem left by prior weighted constructions.
- Originally pursued reinforcement learning and intelligent search to automate gadget discovery; after extensive experimentation the approach failed to produce valid structures, leading the group to pivot to graph decomposition and constructive enumeration — a decision informed directly by the RL failure analysis.
- Developed key theoretical observations on non-bipartiteness requirements for crossing gadgets and tensor-preserving vertex-split techniques to lower pin-degree constraints; characterized pin-ring conditions explaining why all prior lattice embedding attempts failed at the same geometric step.
- Contributed core search and verification algorithms to [GadgetSearch.jl](#): reduced- α tensor certification, Redelmeier lattice-animal enumeration over 5.6 billion candidates ($n \leq 15$), and convex-hull port validation.

LLM Combinatorial Reasoning Benchmark — Prof. Jinguo Liu's Group

HKUST, Hong Kong

RESEARCH CONTRIBUTOR

2025 – 2026

- Motivated by the lack of any benchmark targeting LLM reasoning on combinatorial optimization: the [problem-reductions](#) library (290+ reduction rules) naturally accumulates instances with subtle logical bugs — not syntax errors, but combinatorial reasoning failures in reduction proofs.
- Designed and built the evaluation harness for [problem-reductions-benchmark](#), modeling the pipeline after SWE-bench: automated extraction of buggy reduction instances, structured LLM prompting, and independent certificate verification via the `pred CLI`.
- Defined the primary metric (bugs/Ktok) enabling fair model-agnostic comparison across pricing tiers.

Projects

F5-TTS: Few-Shot Mandarin Voice Cloning with LoRA Adaptation

HKUST

COURSE PROJECT

2025

- Fine-tuned F5-TTS for Mandarin child voice cloning by adding a LoRA adapter, reducing adaptation cost while preserving generation quality.
- Built end-to-end pipeline: Whisper-large-v2 transcription of 220 audio files, hyperparameter sweep over diffusion steps and classifier-free guidance strength, and a multi-metric evaluation suite (PESQ, STOI, DNSMOS, CSIG/CBAK/COVL).

CausVid: Reproducing Causal Video Diffusion Distillation (CVPR 2025)

HKUST

COURSE PROJECT

2025

- Reproduced the asymmetric distillation framework from “From Slow Bidirectional to Fast Autoregressive Video Diffusion Models”, achieving autoregressive streaming generation via Distribution Matching Distillation.
- Validated key results: latency reduction from 219s to ~ 1.3 s initial frame with 9.4 FPS streaming; explored zero-shot video-to-video transfer as an extension.

Education

The Hong Kong University of Science and Technology
B.ENG. IN ARTIFICIAL INTELLIGENCE

Hong Kong
Sep. 2024 – Present

- Relevant Coursework: Algorithms & Data Structures, Linear Algebra, Probability Theory, Machine Learning, Neural Networks, Quantum Computing

Exchange & Industry Experience

UC Berkeley Global Access (BGA) Program
VISITING STUDENT

Berkeley, California
Aug. 2026 – Dec. 2026

- Selected by HKUST for semester exchange; engaging with research groups in AI, robotics, and machine learning at Berkeley.

Hikvision (Hangzhou Headquarters)
SOLUTIONS ENGINEER INTERN

Hangzhou, China
Jun. 2026 – Jul. 2026

- Participated in pre-sales technical solution drafting for smart-city AI projects; gained exposure to production-grade deep-learning pipeline requirements.

Technical Skills

Languages	Python (proficient), Julia (working), Rust (reading), Git, $\LaTeX$
ML & AI	PyTorch, Transformers, LoRA fine-tuning, Diffusion Models, LLM evaluation & prompting
Algorithms & Theory	Combinatorial optimization, NP-hard reductions, Tensor networks, Graph algorithms, MIS / QUBO formulations
Research Interests	LLM reasoning, Agentic systems, Combinatorial problem-solving, Embodied AI